



JING-ZHANG CIVIC SERVICE RELAY

Urban multi-agent collaboration testbed · AI-native urban planning



01 / AI-NATIVE URBAN PLANNING

ONE OPEN TEST SPINE, THREE AI STRESS-TEST GROUNDS

CONCEPT · PROVISIONAL BOUNDARY · SYNTHETIC OFFLINE TEST



NIGHT SUPPORT STATION
resilience / fault injection / takeover

SHARED CARE COURTYARD
graph constraints / parallel critics

ALL-HOURS CIVIC SERVICE HALL
multimodal / receipt replay / appeal

01 TASK GRAPH

LivingTaskGraph · steps / dependencies / time windows

02 STATE + SOLVER

CapacityTwin + constraints → two options

03 PARALLEL CRITICS

equity / safety / data → pass-warn-block

04 BOUNDED REVISION + HUMAN GATE

one revision max → approve / degrade / decline

05 RECEIPT + REPLAY

LifeCapacityReceipt · 54 records / 108 counterfactuals



THREE SCALES × FIVE AI LAYERS

Regional interface, open test spine and three AI laboratories



02 / SPATIAL STRUCTURE

THREE SCALES OF CIVIC SERVICE RELAY

CONCEPT · PROVISIONAL BOUNDARY · SYNTHETIC OFFLINE TEST

01

43.6 km² STRATEGIC STUDY

Regional interfaces connect models, compute, data, tools, talent and open scenarios with independent evaluation.



02

c.11.4 km² OVERALL DESIGN

One open test spine reuses task graphs, service-state models, fixed protocols and a human responsibility gate.



03

368.4 ha THREE KEY AREAS

Three AI laboratories expose resilience, constraint solving and trustworthy interaction in public space.



Areas use current provisional geometry and must be recalculated when official polygons arrive.

LEGEND: TEAL=VERIFIED · RED=STOP/HUMAN · YELLOW=PROVISIONAL | SOURCE: PACKAGE DATA + SUBMITTED GEOJSON | BOUNDARY: PROVISIONAL, NOT OFFICIAL



THREE AI STRESS-TEST GROUNDS

Resilience · constraint solving · trustworthy interaction



THREE KEY AREAS, THREE TECHNICAL-SPATIAL PROTOTYPES

CONCEPT · PROVISIONAL BOUNDARY · SYNTHETIC OFFLINE TEST

01 NIGHT SUPPORT STATION

PLAN / 平面

state board → edge degrade → takeover → receipt



SECTION / 剖面

canopy + safety light + fault-injection area

OPERATION / 运行

NETWORK / STALE-STATE INJECTION

expose inputs, blocks, takeover, version and metrics

02 SHARED CARE COURTYARD

PLAN / 平面

task graph → solver → three parallel critics



SECTION / 剖面

shared floor + time wall + quiet case room

OPERATION / 运行

SCARCITY / SCHEDULE INJECTION

expose inputs, blocks, takeover, version and metrics

03 ALL-HOURS CIVIC SERVICE HALL

PLAN / 平面

multimodal input → ambiguity → replay + appeal



SECTION / 剖面

porous hall + receipt query + staffed window

OPERATION / 运行

FALSE-AVAILABILITY / LOW-CONFIDENCE

expose inputs, blocks, takeover, version and metrics



FIVE-STEP RESPONSIBILITY CHAIN

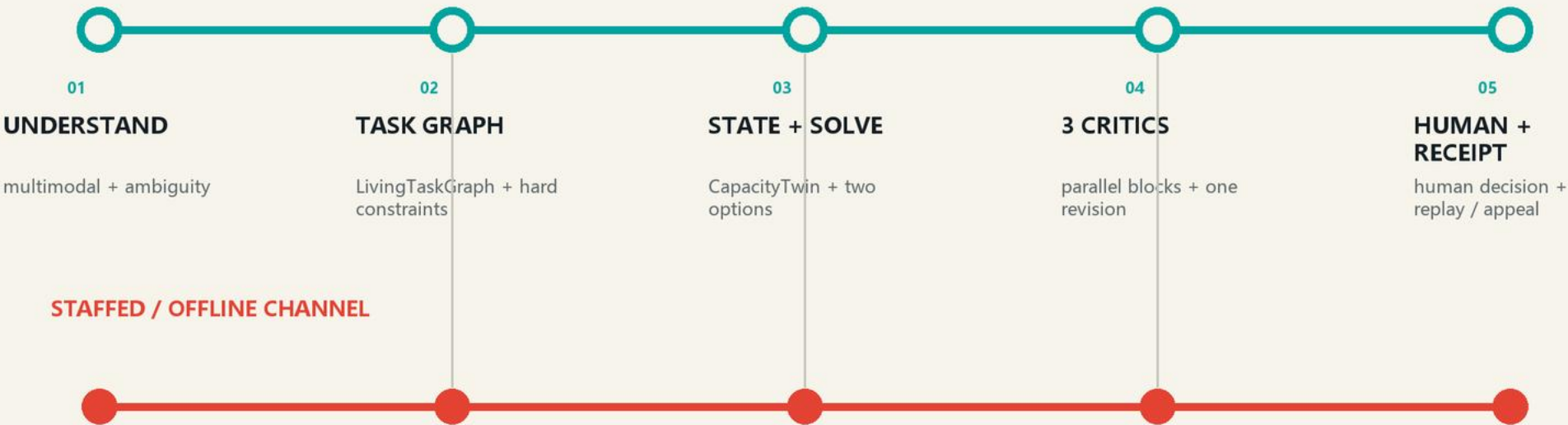
Task graph, state solving, critics, human gate and replay



ONE EVERYDAY NEED, FIVE TECHNICAL RESPONSIBILITY STEPS

CONCEPT · PROVISIONAL BOUNDARY · SYNTHETIC OFFLINE TEST

AI + SPACE



STAFFED / OFFLINE CHANNEL

PHONE / PAPER / HUMAN WINDOW REMAIN PARALLEL



SIX AGENTS + TECHNICAL LEVERAGE

Four high-ROI insertion points · 54 records · 108 counterfactuals



SIX AGENT ROLES + AI LEVERAGE

CONCEPT · PROVISIONAL BOUNDARY · SYNTHETIC OFFLINE TEST

TASK MODELING AGENT

multimodal → task graph / missing fields

SERVICE-SOLVING AGENT

task + state → two options / no solution

EQUITY REVIEW AGENT

comparable cases → gap / pass-warn-block

SAFETY REVIEW AGENT

stale-outage-no staff → degrade / stop

DATA PROTECTION AGENT

fields + purpose + retention → minimum set

RECEIPT REVIEW AGENT

events + human decision → receipt / replay

SAFE RESOLUTION GAIN

+38.9 pp

UNSAFE RELEASE

7 → 0

TAKEOVER TIME SAVED

-147.9 s

RECEIPT GAIN

+33.3 pp

18 fixtures × 3 variants = 54 records
18 fixtures × 6 identities = 108 counterfactuals

AI-08 identity gap 0.0 pp · assigned gap confounded · AI-10 field result unknown